Macroeconomics

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DATE PLACEHOLDER

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1 Fundamental Concepts

Economics is the social science that studies the production, distribution, and consumption of goods and services. At its core, economics addresses the fundamental problem of **scarcity**: while resources such as labor, capital, and natural resources are limited, human wants are essentially unlimited. This creates the central challenge that all economies must navigate—how to allocate scarce resources among competing uses.

Macroeconomics is the branch of economics that studies the economy as a whole, including phenomena such as economic growth, inflation, unemployment, and national income. Unlike microeconomics, which focuses on individual behavior, macroeconomics examines aggregate outcomes and the policies that influence them.

Core Concept 1.1: Factors of Production

The production of goods and services requires four essential factors of production:

(a) **Land** encompasses all natural inputs used in production, including agricultural land, mineral deposits, water resources, and the physical space upon which production facilities are constructed.

(b) **Labor** represents the human effort devoted to production, encompassing both physical and mental work. Labor includes the services of workers across all sectors, from agricultural laborers harvesting crops to knowledge workers developing software.

(c) **Capital** refers to goods that are produced and then used to produce other goods and services. Unlike consumer goods that satisfy wants directly, capital goods facilitate future production. Examples include manufacturing facilities, machinery, transportation equipment, and tools.

(d) **Entrepreneurship** represents the creative effort of individuals who combine the other three factors of production to create goods and services. Entrepreneurs bear risk, organize production, and drive innovation.

Core Concept 1.2: Opportunity Cost

When making economic decisions, individuals and societies must choose among mutually exclusive alternatives. The **opportunity cost** of a choice represents the value of the best alternative forgone. This concept is fundamental to rational decision-making because choosing one option necessarily means sacrificing another.

Core Concept 1.3: Production Possibilities Curve

The **Production Possibilities Curve (PPC)** illustrates the maximum combinations of two goods that an economy can produce given its available resources and technology.

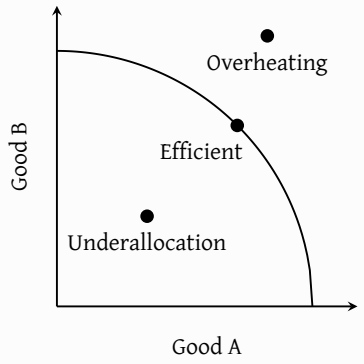

Figure 1: Production possibilities curve

(a) Points **on the curve** represent efficient production where all resources are fully utilized.

(b) Points **inside the curve** indicate underallocation of resources

(c) Points **outside the curve**—although appearing unattainable—can occur temporarily during periods of economic overheating when resources are used beyond their sustainable capacity. However, this state is not sustainable and typically leads to inflation, resource depletion, or economic crisis.

The **shape of the PPC** reveals important information about opportunity costs. A concave curve (Figure 1) indicates increasing opportunity costs, meaning that producing additional units of one good requires sacrificing increasingly larger quantities of the other good. This typically reflects the fact that resources are not equally adaptable to producing both goods.

Shifts in the PPC represent economic growth or decline. **Outward shifts indicate economic growth**, resulting from increases in available resources, improvements in technology, or enhancements in human capital. On the other hand, **inward shifts indicate economic decline**, potentially from natural disasters, war, or resource depletion.

Core Concept 1.4: Comparative and Absolute Advantage

An economy has an **absolute advantage** when it can produce more of a good than another economy using the same resources. **Comparative advantage** is when an economy can produce a good at a lower opportunity cost than another economy.

It is impossible for one economy to have a comparative advantage in both goods simultaneously. This principle, known as the **law of comparative advantage**, implies that all economies can benefit from trade by specializing in goods where they hold a comparative advantage. The gains from trade arise from this specialization and the subsequent exchange of goods.

Core Concept 1.5: Demand

Demand represents the quantity of a good that consumers are willing and able to purchase at various prices during a given time period. The **Law of Demand** states an inverse relationship between price and quantity demanded: as prices rise, quantity demanded falls, and vice versa. This relationship reflects both the **substitution effect** (*consumers shift to cheaper alternatives*) and the **income effect** (*higher prices reduce purchasing power*).

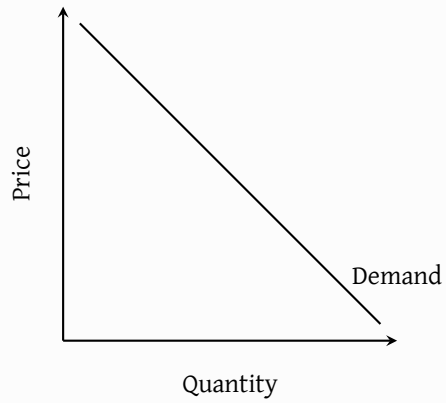

Figure 2: Demand curve

The determinants of demand, factors that can cause the entire demand curve to shift, include:

(a) **Consumer income**: Normal goods experience increased demand as income rises, while inferior goods experience decreased demand.

(b) **The number of buyers** in the market.

(c) **The prices of substitute and complementary goods**.

(d) **Consumer expectations** about future prices.

(e) **Consumer tastes and preferences**.

Core Concept 1.6: Supply

Supply represents the quantity of a good that producers are willing and able to offer for sale at various prices. The **Law of Supply** indicates a direct relationship between price and quantity supplied: higher prices incentivize producers to increase production.

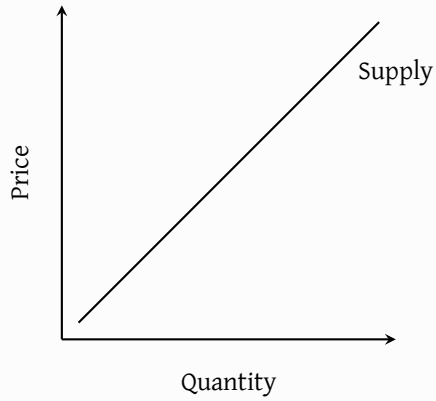

Figure 3: Supply curve

Determinants of supply include:

(a) **Production costs**, including input prices and technology.

(b) **The prices of related goods** firms could produce.

(c) **Expectations about future prices**.

(d) **The number of sellers** in the market.

(e) **Government policies**, such as taxes, subsidies, and regulations.

Core Concept 1.7: Market Equilibrium

Market Equilibrium occurs where the supply and demand curves intersect. At this point, the quantity supplied equals the quantity demanded, and the market clears. The equilibrium price and quantity are determined automatically through the price mechanism without external intervention. Correspondingly, adjustments in equilibrium price occur when supply and demand curve shifts.

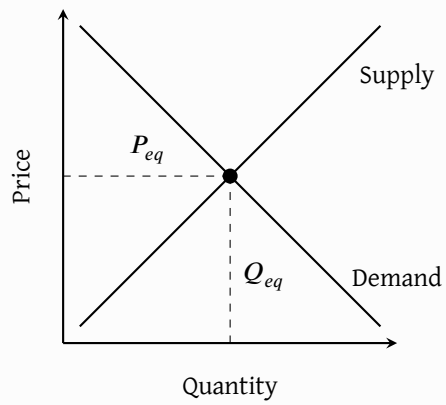

Figure 4: Market equilibrium

Core Concept 1.8: Circular Flow of Income

The **Circular Flow of Income** illustrates how goods, services, and money flow between **households** and **firms** in an economy. Firms produce goods and services using factors of production provided by households and, in exchange, pay households wages, rent, interest, and profits. Households then use this income to purchase goods and services from firms, completing the circular flow.

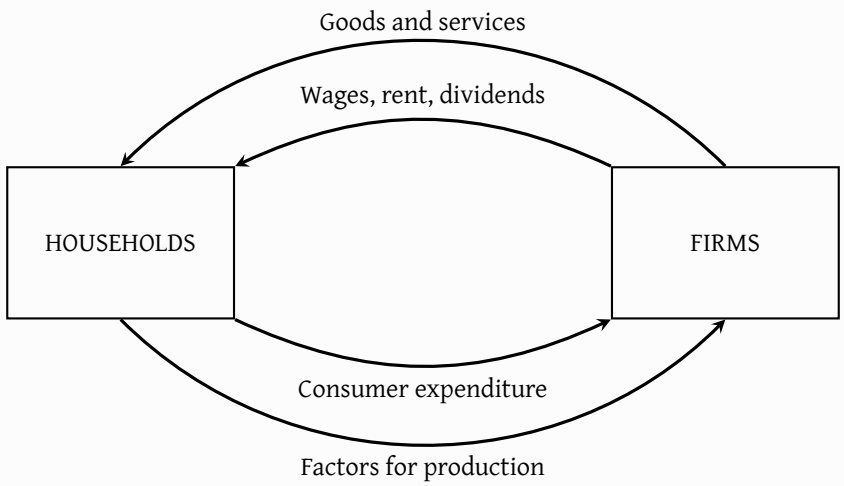

Figure 5: Circular flow of income

This model (Figure 5) illustrates the interdependence between sectors of the economy and forms the basis for measuring aggregate economic activity through Gross Domestic Product (Core Concept 2.1).

2 Economic Indicators

Macroeconomic analysis relies on various **economic indicators** to assess the health and performance of an economy. These indicators provide insight into economic trends, business cycles, and the effectiveness of policy interventions.

2.1 Gross Domestic Product

Core Concept 2.1: Gross Domestic Product

Gross Domestic Product (GDP) represents the total market value of all **final** goods and services produced within a country's borders during a specific time period, typically one year or one quarter. GDP does not include the following items.

(a) **Transfer payments**: Government payments not tied to current production, such as Social Security.

(b) **Intermediate goods**: Which are counted in final goods to avoid double-counting.

(c) **Financial transactions**, like stock purchases.

(d) **Goods produced in previous periods**, such as existing home sales.

In general, GDP serves as the primary measure of economic output and standard of living.

Core Concept 2.2: Economic Cycles

The **business cycle** represents fluctuations in economic activity around the long-term growth trend. The cycle comprises four phases: expansion (*real GDP rising*), peak (*the highest point of economic activity*), recession (*real GDP falling*), and trough (*the lowest point*).

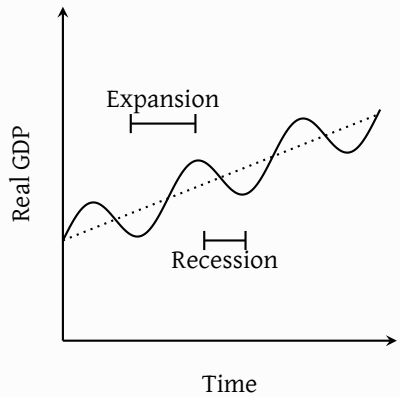

Figure 6: Business cycles

As shown in Figure 6, the **output gap** measures the difference between actual GDP and potential GDP (*the level of GDP that would exist if all resources were fully utilized*). A positive output gap indicates the economy is operating above its sustainable capacity, while a negative output gap indicates underutilization of resources.

Economists typically define a **recession** as two consecutive quarters of negative real GDP growth, though the National Bureau of Economic Research uses more comprehensive criteria.

GDP can be calculated through multiple approach that would yield equivalent results

Core Concept 2.3: Calculating GDP (*Expenditures Approach*)

The **expenditures approach**—the most common approach—sums all spending on final goods and services.

$$\mathbf{GDP = C + I + G + X - M}$$

Here,

(a) **C** represents household consumption (*spending on final goods and services*).

(b) $\mathbf{I}_g$ represents gross investment (*spending on capital goods and inventory, excluding intermediate goods*).

(c) **G** represents government purchases (*government spending on goods and services, excluding transfer payments*).

(d) **X** represents exports.

(e) **M** represents imports.

Core Concept 2.4: Net Domestic Product (NDP)

Net domestic product (NDP) is GDP but accounts for the depreciation of capital goods. Its calculation takes the same method as the expenditures approach (Core Concept 2.3), but substitutes gross investment with net investment. **Net investment ($\mathbf{I}_n$)** is gross investment minus depreciation (*the decrease in value of capital goods over time*).

While GDP is the headline figure mainstream media usually follows, NDP is a more honest figure because it measures only the investment used for additional growth on the existing captial base, excluding the portion that goes into simply replacing worn down captial.

Core Concept 2.5: Calculating GDP (*Income Approach*)

The **income approach** calculates GDP by summing all incomes earned in production. This includes wages and salaries, interest income, rental income, profits (*including proprietor income and corporate profits*), and statistical adjustments (*depreciation, sales taxes, and net foreign factor income*). The equality of expenditure and income approaches reflects the circular flow of income.

$$\text{GDP} = \text{Wages} + \text{Interest} + \text{Rental Income} + \text{Profits} + \text{Statistical Adjustments}$$

Core Concept 2.6: Calculating GDP (Value Added Approach)

The **value added approach** calculates GDP by summing the value added at each stage of production. Value added equals the value of a firm's output minus the value of intermediate goods it purchases. This approach avoids double-counting by measuring only the incremental value contributed at each production stage.

$$\text{GDP} = \sum \text{Value Added at Each Stage}$$

For example, consider bread production: a farmer sells wheat for \$1, a mill sells flour for \$2, and a bakery sells bread for \$3. The total value added is \$1 + (\$2 - \$1) + (\$3 - \$2) = \$3, which equals the final selling price.

The three methods of calculating GDP—expenditure, income, and value-added—are essentially three ways of counting the same dollar as it travels through the economy's circular flow. The expenditure approach catches that dollar at the cash register when a final good is purchased, the income approach tracks it as it is paid out to workers and business owners, and the value-added approach measures it as it is "built" piece-by-piece at every stage of production. Because every transaction is a closed loop where one person's spending is another person's income and a direct reflection of the new value created, all three paths inevitably lead to the exact same total: the market value of everything produced in a year—or GDP.

Core Concept 2.7: Limitations of GDP

Despite its widespread use, GDP has important limitations as a measure of economic well-being.

(a) GDP excludes **non-market transactions** and **underground economic activity**, such as household production, volunteer work, and illegal transactions.

(b) GDP does not capture changes in **environmental quality**, **income inequality**, or **social conditions**. A country with high GDP may also have significant pollution, income disparity, or social unrest.

(c) GDP does not account for **depreciation**, which overstates true economic welfare. NDP, as outlined in Core Concept 2.4, accounts for this inaccuracy.

Core Concept 2.8: Economic Growth

Economic Growth refers to a sustained increase in real GDP per capita (*per person*) over time, representing the expansion of an economy's productive capacity and living standards.

Long-run economic growth is represented by rightward shifts in the long-run aggregate supply curve (Core Concept 3.4) and outward shifts in the PPC (Core Concept 1.3). The aggregate production function expresses total output as a function of capital stock and human capital.

$$Y = A \cdot f(L, K, H, N)$$

Here,

(a) Y: Total Real Output (Real GDP).

(b) A: Total Factor Productivity (TFP). This represents technology, efficiency, and the "know-how" of the economy.

(c) L: Labor. The total number of workers or hours worked.

(d) K: Physical Capital. Tools, machinery, factories, and infrastructure.

(e) H: Human Capital. The skills, education, and experience of the workforce.

(f) N: Natural Resources. Land, minerals, and raw materials.

Labor productivity (*output per worker*) is a key determinant of economic growth. Factors driving growth include:

(a) **Capital accumulation**: Investment in machinery and infrastructure

(b) **Human capital development**: Education and skills training

(c) **Technological advancement**

(d) **Increases in the labor force participation rate** (Core Concept 2.9)

2.2 Labor Market

The **labor market** provides crucial information about employment, unemployment, and the economy's productive capacity.

Core Concept 2.9: Labor Force Participation Rate

The **labor force participation rate** (LFPR) measures the proportion of the working-age population actively in the labor market. It is calculated by the following equation.

$$\text{LFPR} = \left(\frac{\text{Labor Force}}{\text{Civilian Non-institutional Population}} \right) \times 100$$

The **labor force** is defined as the sum of all individuals who are either currently working or actively seeking work, excluding discouraged workers who have stopped seeking employment. The **civilian noninstitutional population** represents the "pool" individuals who are legally and physically able to participate in the economy.

Core Concept 2.10: Unemployment Rate (U-3)

The **unemployment rate** (*or also called the U-3 rate*) measures the proportion of unemployed persons to the total labor force. This is a major headline statistic that all news outlets, economists, investors, etc. follow.

$$\text{Unemployment Rate} = \left(\frac{\text{Number of Unemployed}}{\text{Labor Force}} \right) \times 100$$

For many advanced economies, including the U.S., and unemployment rate of 3.5% to 4.5% is considered the "sweet spot" that governments strive for.

Core Concept 2.11: Types of Unemployment

Unemployment can be further categorized into the following baskets.

(a) **Frictional unemployment** occurs when workers are between jobs or searching for new employment. This type of unemployment is inevitable and even desirable in a dynamic economy, as it reflects workers seeking better matches.

(b) **Structural unemployment** arises from mismatches between worker skills and job requirements, often due to technological change or shifts in the industrial composition of the economy. This unemployment persists even when the economy is at full employment.

(c) **Cyclical unemployment** results from economic recessions and fluctuations in aggregate demand. When the economy contracts, employers lay off workers, increasing cyclical unemployment.

The **natural rate of unemployment (NRU)** equals the sum of frictional and structural unemployment. It represents the unemployment rate when the economy is at full employment, consistent with stable inflation.

Core Concept 2.12: Underemployment (U-6)

The **underemployment rate** (*also known as the U-6 rate*) represents the proportion of unemployed and underemployed persons to the total labor force. Underemployed persons consists of **marginally attached workers** (people who want a job and have looked for one in the last 12 months, but not in the last 4 weeks) and **part-time workers for economic reasons** (people who want full-time work (35+ hours) but are working part-time because their hours were cut or they cannot find a full-time position).

$$\text{Underemployment Rate} = \left(\frac{\text{Unemployed} + \text{Underemployed}}{\text{Labor Force} + \text{Marginally Attached}}\right) \times 100$$

Many economists argue that the U-6 rate provides a much more accurate picture of "labor underutilization" because it includes people who are technically working but are struggling.

Core Concept 2.13: Other Indicators

The following are other important labor market indicators.

(a) **Nonfarm Payrolls**: The change in the number of employed workers excluding farm workers, private household employees, and nonprofit employees. This is the primary gauge of labor market strength.

(b) **Quits Rate**: The number of employees who voluntarily left their jobs during the reporting period, divided by total employment. A higher quits rate indicates worker confidence in finding better opportunities, signaling a strong labor market.

(c) **Jobs Opening Rate**: The number of job openings at the end of the reporting period, divided by the sum of employment and job openings. This measures employer demand for labor and is a leading indicator of future hiring and wage growth.

(d) **Average Hourly Earnings**: The average gross hourly wages paid to nonfarm private employees, excluding government workers. Changes in this measure signal wage inflation pressures and are watched closely as a component of labor costs.

(e) **Initial Claims**: The number of new unemployment insurance claims filed by recently unemployed workers. Lower initial claims indicate a stronger labor market, while rising claims may signal economic weakening.

2.3 Inflation

Core Concept 2.14: Inflation

Inflation represents a sustained increase in the general price level, equivalent to a decrease in the purchasing power of money.

Inflation is primarily kept track of by the following two statistics.

Core Concept 2.15: Price Indices

The **Consumer Price Index (CPI)** is an inflationary measure which tracks the cost of a market basket of goods and services purchased by a typical consumer. This figure is a highly tracked statistic by new outlets and investors.

$$\text{CPI} = \frac{\text{Cost of market basket in current prices}}{\text{Cost of market basket in base period prices}} \times 100$$

The **Producer Price Index (PPI)** measures the average change over time in selling prices received by domestic producers for their output. Unlike the CPI, which tracks prices from the consumer perspective, the PPI reflects inflation from the producer's viewpoint. The PPI serves as a **leading indicator** of consumer inflation because cost changes at the producer level are eventually passed downstream to consumers. While less prominent in popular media than the CPI, the PPI remains a closely monitored inflation gauge among economists and policymakers.

Both the CPI and PPI have **core** versions—**core CPI** and **core PPI**—which exclude food and energy prices from their calculations. These components are omitted because they exhibit high volatility driven by natural factors (*such as weather and crop yields*) and geopolitical events (*such as oil supply disruptions*). By excluding these volatile elements, core measures provide smoother, more stable trends compared to their raw counterparts.

For the U.S., these figures are maintained by the Bureau of Labor Statistics (BLS) and are the main inflation figures used by the U.S. government for administrative and legal purposes.

Core Concept 2.16: Personal Consumption Expenditures

The **Personal Consumption Expenditures (PCE)** index is another inflation index that captures a broader range of consumer spending and accounts for substitution effects. This inflation measure is also another highly tracked statistic.

Like price indices, the PCE index also has a core version for the same reasons.

For the U.S. this figure is maintained by Bureau of Economic Analysis and is the preferred inflation gauge for the Federal Reserve in the context of setting the Federal Funds Rate (Core Concept 5.7).

Core Concept 2.17: Inflation Rate

The **inflation rate** equals the percentage change in the price level:

$$\text{Inflation Rate} = \frac{\text{Current Inflation Figure} - \text{Previous Inflation Figure}}{\text{Previous Inflation Figure}} \times 100$$

Economists and news outlets typically prioritize the **Year-over-Year (YoY)** rate because it provides a more stable picture of long-term trends by filtering out "noise" and seasonal fluctuations that often distort the

Month-over-Month (MoM) data.

$$\text{MoM \%} = \left(\frac{\text{Value}_{currentmonth} - \text{Value}_{previousmonth}}{\text{Value}_{previousmonth}} \right) \times 100$$

$$\text{YoY \%} = \left(\frac{\text{Value}_{currentmonth} - \text{Value}_{samemonthlastyear}}{\text{Value}_{samemonthlastyear}} \right) \times 100$$

Core Concept 2.18: Nominal versus Real Variables

Nominal variables are measured in current dollars and are not adjusted for inflation. Examples include **nominal GDP** (*measured in current prices*) and nominal interest rates (*the stated rate without inflation adjustment*).

Real variables are adjusted for inflation and reflect actual purchasing power. **Real GDP** uses constant base-year prices. For instance, the following is the equation for **real interest rate**.

$$\text{Real Interest Rate} = \text{Nominal Interest Rate} - \text{Inflation Rate}$$

However, most real interest rates reflect the expected future shift in inflation rather than realized inflation, thus the more accurate relationship is:

$$\text{Real Interest Rate} \approx \text{Nominal Interest Rate} - \text{Expected Inflation}$$

This distinction is crucial for economic decision-making, as borrowers and lenders care about real costs and returns. Lenders prefer to increase their lending rates when inflation is expected to rise, while borrowers prefer to delay borrowing when inflation is expected to fall. Consequently, inflation expectations serve as a focal point in determining the quantitative terms of debt contracts and financial arrangements. Many contracts, including loans, bonds, and wage agreements, are written in nominal terms, meaning their real value depends heavily on the inflation rate over their duration.

Unexpected inflation redistributes wealth from creditors to debtors. If a lender sets a nominal interest rate expecting low inflation but actual inflation turns out to be high, the real return becomes negative, benefiting the borrower at the lender's expense. This uncertainty complicates financial planning and investment decisions.

The **GDP deflator** provides another measure of the price level, representing how much of nominal GDP growth is just due to inflation.

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

2.4 Housing

Housing indicators provide valuable insights into the broader economic outlook. These statistics are closely watched because of housing's unique role in the economy.

Core Concept 2.19: Housing and Its Effects on the Economy

Although housing represents only approximately 3% of the economy, it serves as a powerful economic indicator for several reasons. Housing purchases typically involve 30-year mortgages, making housing finance deeply intertwined with the financial system. Additionally, housing purchases trigger purchases of complementary goods, including furniture, appliances, and home improvement materials.

The strong correlation between housing activity and GDP makes housing indicators valuable for predicting economic trends. The 2008 financial crisis demonstrated how housing market troubles can cascade through the entire financial system, particularly through mortgage-backed securities and derivatives tied to residential real estate.

Core Concept 2.20: Housing Indicators

The following are the major housing indicators followed by economists

(a) **Housing Starts**: The number of new residential construction projects that have begun during the reporting period. This is one of the first indicators to decline at the onset of a recession and one of the last to recover, making it a reliable leading indicator of economic trends.

(b) **New Housing Sales**: The number of newly constructed homes with a signed sales contract. New home sales represent a smaller portion of the market than existing sales but are considered a leading indicator because they reflect consumer confidence in the economy and housing market.

(c) **Existing Housing Sales**: The number of previously constructed homes with a completed transaction. Existing home sales make up the majority of housing market activity and reflect the supply side of the market, including homeowner decisions to sell.

(d) **Mortgage Rates**: The interest rates charged on home loans, most commonly tracked for 30-year and 15-year fixed-rate mortgages. Mortgage rates directly affect housing affordability and demand, with lower rates typically stimulating home purchases and higher rates slowing the market.

2.5 Sentiment Indicators

Core Concept 2.21: Economic Sentiment

Economic Sentiment is a composite indicator that measures the overall "mood" or optimism of an economy. It is a **leading indicator**, meaning it often predicts changes in GDP before they occur.

Sentiment is typically subdivided into two primary pillars:

- **Business Confidence:** Measures the expectations of firm managers regarding production, order books, and inventory levels. It predicts future Investment (I).
- **Consumer Confidence:** Measures households' plans for major purchases and their outlook on personal finances. It predicts future Consumption (C).

Core Concept 2.22: Key Consumer Sentiment Indicators

University of Michigan Consumer Sentiment: A monthly survey of at least 500 households that measures consumers' attitudes toward current and expected financial conditions, including their assessment of present economic conditions and their expectations for the future. Readings greater than 50 indicate positive consumer sentiment and typically signal stronger future consumer spending.

Conference Board Consumer Confidence: A monthly survey of approximately 3,000 households that assesses consumer attitudes toward business conditions, employment, and income. The index comprises two components: the present situation index (*current economic conditions*) and the expectations index (*short-term outlook for business conditions, employment, and income*).

University of Michigan Inflation Expectations: A survey-based measure of consumers' expectations for inflation over the next one to five years. These expectations directly influence wage demands, pricing decisions, and actual inflation outcomes, making this a closely watched indicator for monetary policy.

Core Concept 2.23: Key Business Sentiment Indicators

ISM Manufacturing PMI: A monthly survey of purchasing managers at over 300 manufacturing firms conducted by the Institute for Supply Management. The PMI summarizes whether business conditions have improved, stayed the same, or worsened. A reading above 50 indicates expansion, while below 50 indicates contraction. The PMI is one of the most reliable leading indicators of economic activity.

NFIB Small Business Optimism: A monthly survey of small business owners conducted by the National Federation of Independent Business that measures their expectations for the economy, hiring plans, capital expenditure plans, and earnings. Small businesses are significant employers, making this index a useful gauge of labor market and economic conditions.

2.6 Economic Calendar and Indicator Releases

Economic data releases follow predictable schedules, allowing markets to anticipate and react to new information. The following table organizes key indicators from each category.

Category	Indicator	
GDP	Nominal GDP	Nominal GDP per capita
	Real GDP	Real GDP per capita
Inflation	CPI	PPI
	PCE	
Labor Market	U-3 Rate	U-6 Rate
	Participation Rate	Nonfarm Payrolls
	Quits Rate	Jobs Opening Rate
	Average Hourly Earnings	Initial Claims
Housing	Housing Starts	New Housing Sales
	Existing Housing Sales	30-Year Mortgage Rate
	15-Year Mortgage Rate	
Sentiment	ISM Manufacturing PMI	UMich Consumer Sentiment
	CB Consumer Confidence	UMich Inflation Expectations
	NFIB Small Business Optimism	
Other	VIX	Dollar Index
	Federal Funds Rate	

Table 1: Key economic indicators by category

Core Concept 2.24: World Economic Calendar

The **world economic calendar** is a comprehensive schedule of economic indicator releases (such as GDP, CPI, and employment data) and central bank meetings throughout the year.

It is the primary tool used by investors and policymakers to track **market expectations** versus **actual results**. Discrepancies between the "forecast" and the "actual" data typically drive volatility in global currency and stock markets.

3 Macroeconomic Relationships

The macroeconomy functions through interconnected relationships between aggregate demand, aggregate supply, and various policy variables.

3.1 Aggregate Demand

Core Concept 3.1: Aggregate Demand

Aggregate Demand (AD) represents the total quantity of goods and services demanded in the economy at different price levels.

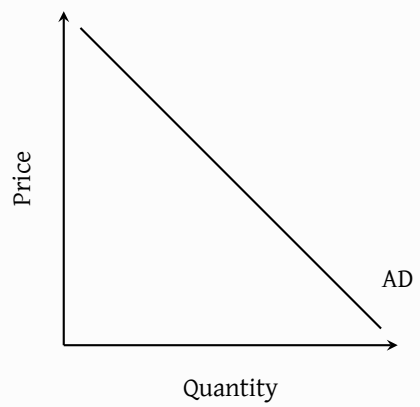

Figure 7: Aggregate demand curve

The AD curve slopes downward for three reasons:

(a) **Wealth effect**: Lower price levels increase real wealth, allowing consumers to purchase more goods and services.

(b) **Interest rate effect**: Lower price levels reduce money demand, causing interest rates to fall, stimulating investment.

(c) **Exchange rate effect**: Lower interest rates cause currency depreciation, making domestic goods cheaper abroad and stimulating exports.

Rightwards Shifts in the AD curve result from changes in the following four components. Correspondingly, decreases shift the AD leftward.

(a) **Consumer spending**: Increases from tax cuts or increased confidence shift AD rightward.

(b) **Investment**: Increases from business optimism or technological innovation shift AD rightward.

(c) **Government spending**: Increases shift AD rightward.

(d) **Net exports**: Increases from foreign economic growth or currency depreciation shift AD rightward.

Core Concept 3.2: The Multiplier Effect

An initial change in spending creates a larger final change in GDP through the multiplier effect. To understand why this is, we introduce two new concepts: the **Marginal Propensity to Consume or MPC** (the fraction of additional disposable income that households spend) and the **Marginal Propensity to Save (MPS)** (the fraction of additional household income that households save). Thus,

$$\text{MPC} + \text{MPS} = 1$$

The **spending multiplier** effect operates through successive rounds of spending. When one person spends income, the recipient receives income and spends a portion (*the MPC*), continuing until leakages (*saving, taxes, imports*) absorb the remaining income.

$$\text{Spending Multiplier} = \frac{1}{1 - \text{MPC}} = \frac{1}{\text{MPS}}$$

The **tax multiplier** works in the same manner but with a smaller initial impact because a tax change first affects disposable income rather than direct aggregate demand. When taxes are cut, consumers do not spend the entire amount; they save a portion (*the MPS*) immediately. Consequently, the first round of the spending cycle is "diluted" by this initial leakage, making the tax multiplier's overall effect less powerful than that of the spending multiplier. The negative sign in the following equation for the tax multiplier reflects that tax increases reduce disposable income and spending.

$$\text{Tax Multiplier} = -\frac{\text{MPC}}{1 - \text{MPC}}$$

3.2 Aggregate Supply

Core Concept 3.3: Short-Run Aggregate Supply

Short-Run Aggregate Supply (SRAS) represents the total output firms are willing to produce at different price levels in the short run, when some input costs (*particularly wages*) are fixed.

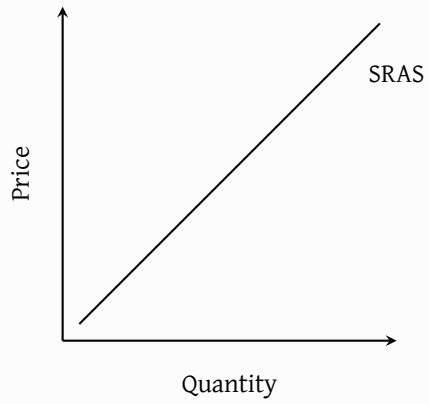

Figure 8: Short-run aggregate supply

The SRAS curve slopes upward because higher price levels increase profitability when input costs are **sticky**. Firms respond to higher prices by increasing production to capture additional profits in the near future until production costs adjust.

The SRAS curve illustrates the short-run trade-off between inflation and unemployment. Points higher on the curve indicate economic boom conditions (*low unemployment, high inflation*), while points lower on the curve indicate recessionary conditions (*high unemployment, low inflation*).

SRAS shifts result from changes in productivity, input costs, labor market conditions, and taxes or subsidies.

Core Concept 3.4: Long-Run Aggregate Supply

Long-Run Aggregate Supply (LRAS) represents the economy's potential output when all input costs can fully adjust to price changes. In the long run, output is independent of the price level and equals potential GDP.

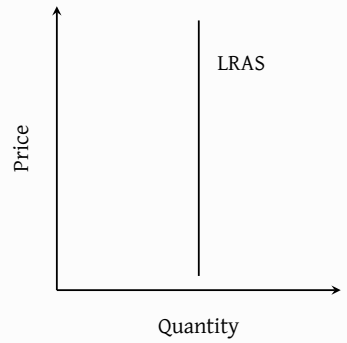

Figure 9: Long-run aggregate supply

3.3 Economic Theory

Core Concept 3.5: Types of Inflation

Demand-pull inflation results from AD increasing beyond potential output, pushing the economy above its LRAS.

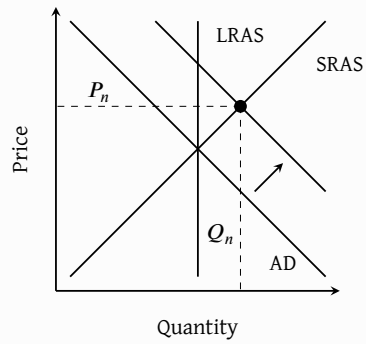

Figure 10: Demand pull inflation

Cost-push inflation results from SRAS decreasing due to supply shocks, reducing output and raising prices simultaneously.

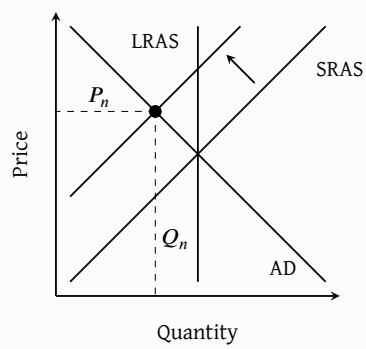

Figure 11: Supply push inflation

Core Concept 3.6: Long-Run Self-Adjustment & Equilibrium

Without government intervention, the economy naturally adjusts to long-run equilibrium through wage and price flexibility. When output falls below potential (*recessionary gap*), unemployment rises, wages decline, and production costs fall, shifting SRAS rightward until output returns to potential. When output exceeds potential (*inflationary gap*), upward pressure on wages increases costs, shifting SRAS leftward.

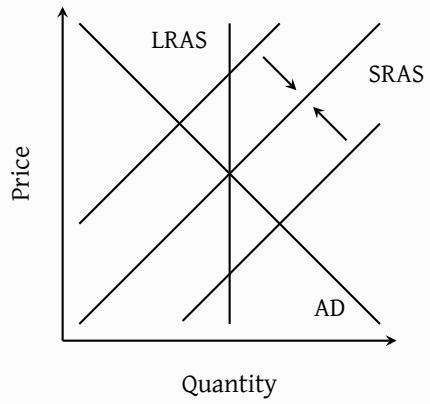

Figure 12: Long-run self-adjustment or SRAS

This self-adjustment process explains why classical economists emphasized the economy's natural tendency toward full employment (U-3 Rate = NRU), though the adjustment process may be slow and painful.

Core Concept 3.7: The Phillips Curve

The **Phillips Curve** illustrates the short-run inverse relationship between unemployment and inflation. Lower unemployment corresponds to higher inflation, reflecting the short-run trade-off between economic activity and price stability.

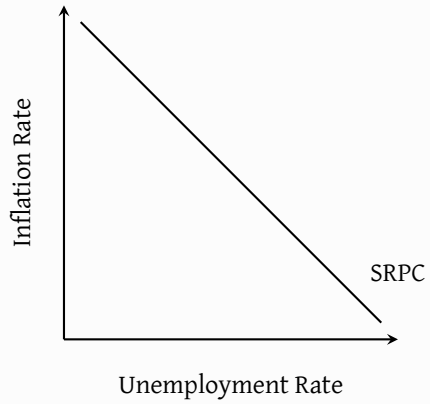

Figure 13: Short-run phillips curve

Movements along the SRAS correspond to movements along the Short-Run Phillips Curve (SRPC). Rightward shifts in SRAS shift the SRPC leftward (*better outcomes for both unemployment and inflation*). However, a leftward shift in the SRAS (*supply push inflation*) causes a rightward shift in the SRPC, resulting in a phenomenon known as **stagflation**—a period characterized by simultaneous increases in both unemployment and inflation.

In the long run, the economy returns to the natural rate of unemployment regardless of inflation, making the Long-Run Phillips Curve vertical.

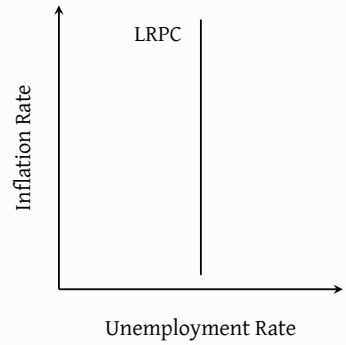

Figure 14: Long-run phillips curve

Core Concept 3.8: Quantity Theory of Money

The **Quantity Theory of Money** provides a mathematical framework for understanding the long-run causes of inflation. The theory is expressed through the following identity.

$$MV = PQ$$

Here, M represents the **money supply**, V represents **velocity** (*how frequently a unit of money changes hands*), P represents the **price level**, and Q represents the **quantity of real output** (*or real GDP*).

Classical economists assume that both velocity (V) and real output (Q) are relatively stable or determined by non-monetary factors in the long run.

- **Velocity** is determined by spending habits and banking technology.
- **Output** is determined by the economy's productive capacity (*labor, capital, and technology*).

Because V and Q are assumed to be stable, any change in the money supply (M) must result in a proportional change in the price level (P). Therefore, sustained inflation—a persistent rise in P—is interpreted as the direct result of a sustained increase in the money supply.

4 Financial Sector

The financial sector channels savings to investment and facilitates economic transactions.

4.1 Money and the Money Market

Core Concept 4.1: Functions of Money

Money serves the following three essential functions in an economy.

- **Medium of exchange**: Money facilitates transactions by eliminating barter's double coincidence of wants.
- **Store of value**: Money preserves purchasing power for future use.
- **Unit of account**: Money provides a common measure for valuing goods and services.

When hyperinflation erodes money's value, it fails as a store of value and unit of account, disrupting economic activity.

The **opportunity cost of holding money** is the interest that could be earned by holding interest-bearing assets instead. Higher interest rates increase this opportunity cost, reducing money demand because people would much rather earn interest from securities.

Core Concept 4.2: Money Supply (M0, M1, and M2)

Economists categorize the money supply using the following aggregates.

- **M0**: Currency in circulation plus reserves held by banks at the central bank.
- **M1**: The most liquid forms, including currency, checking accounts, and travelers' checks.
- **M2**: Includes M1 plus savings deposits, small time deposits, and money market mutual funds.

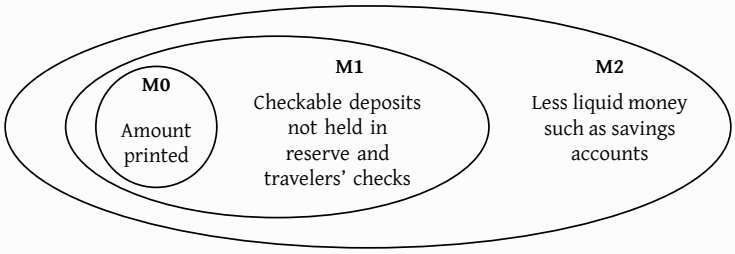

Figure 15: Diagram of money supply

These aggregates reflect different degrees of liquidity, with broader measures (*M2*) capturing more forms of money that can be relatively easily converted to cash.

Core Concept 4.3: The Money Market

The money market determines the equilibrium nominal interest rate through the interaction of money supply and money demand. The money supply curve is vertical (*set by the central bank*), while money demand slopes downward (*higher interest rates increase the opportunity cost of holding money*).

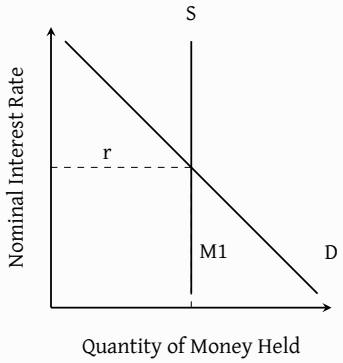

Figure 16: Money market

Factors affecting money demand include price levels (*higher prices require more money for transactions*), national income (*higher income increases transactions*), and technology affecting liquidity.

4.2 Banking System

Core Concept 4.4: Fractional Reserve Banking

Banks operate under **fractional reserve banking**, maintaining only a fraction of deposits as **reserves** while lending the remainder. This reserve to lending ratio is set by the **required reserve ratio (RRR)**, which establishes the minimum fraction of deposits that banks must hold as reserves.

This system creates money through the multiplier effect as shown by the following formulas.

$$\text{money multiplier} = \frac{1}{\text{RRR}}$$

$$\Delta\text{Money Supply} = \text{Excess Reserves} \times \frac{1}{rr}$$

As these loans are spent and redeposited into other banks, the process repeats across the banking system, mathematically expanding a single initial deposit into a much larger total increase in the money supply. We have seen this same pattern with the multiplier effect in the context of MPC and MPS (Core Concept 3.2). With a 20% RRR, an initial deposit of $1,000 can ultimately support $5,000 in total deposits.

Core Concept 4.5: Loanable Funds Supply and Demand

The loanable funds market determines the real interest rate through the supply of funds (*savings*) and demand for funds (*investment*). With higher real interest rates, the supply of loanable funds increases (*encouraging saving*) and the demand decreases (*discouraging borrowing for investment*).

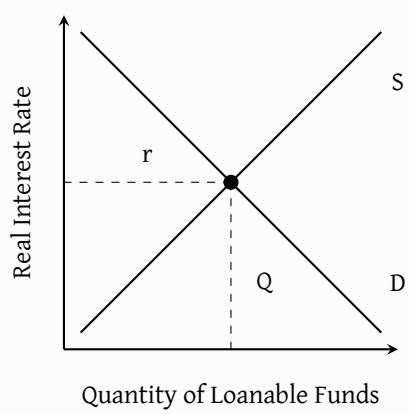

Figure 17: Loanable funds

The position of these curves reflects national saving behavior (S), government borrowing (D), and investment profitability (D) expectations.

Core Concept 4.6: National Savings

National Savings represents total saving in an economy for investment, equaling private savings plus public savings. A **closed economy** national savings is represented by the following equation.

$$I = \text{National Savings} = Y - C - G = \underbrace{(Y - T - C)}_{\text{Private Saving}} + \underbrace{(T - G)}_{\text{Public Saving}}$$

As the above equations shows nicely, private saving is simply household income (Y) after taxes (T) and private consumption (C) and public saving (*or the government saving*) its tax revenue (T) after accounting for its spending (G).

In an **open economy**, the relationship between savings and investment changes because a country is no longer limited to its own internal resources. It can interact with the rest of the world and must consider net capital outflow (NCO).

$$\text{National Savings} = I + NCO$$

Net Capital Outflow (NCO) measures the net flow of funds being invested abroad, calculated by the following equation.

$$\text{NCO} = \text{Domestic purchase of foreign assets} - \text{Foreign purchase of domestic assets}$$

5 Stabilization Policies

Policymakers use fiscal and monetary policy to manage aggregate demand and stabilize the economy.

5.1 Fiscal Policy

Core Concept 5.1: Fiscal Policy

Fiscal policy involves government spending and taxation decisions to influence aggregate demand.

- **Contractionary fiscal policy** (*reducing spending or increasing taxes*) reduces AD to combat inflation.
- **Expansionary fiscal policy** (*increasing spending or decreasing taxes*) stimulates AD to combat recession.

Keep in mind that the initial impact of fiscal policy is multiplied through the multiplier effect (Core Concept 3.2), meaning the ultimate change in GDP exceeds the initial fiscal impulse.

Core Concept 5.2: Automatic Stabilizers

Automatic stabilizers are fiscal policies that automatically counteract economic fluctuations without discretionary action by the government. These stabilizers provide immediate response to economic conditions, avoiding recognition and implementation lags that affect discretionary policy.

Progressive income taxes exemplify this: during expansions, rising incomes push taxpayers into higher brackets, automatically increasing tax revenues and exerting downward pressure on the economy. During recessions, falling incomes reduce tax burdens, providing automatic stimulus.

Core Concept 5.3: Crowding Out and Government Debt

Government **deficit** (*annual spending minus revenue*) and **debt** (*accumulated deficits*) represent the fiscal stance. Deficit spending can stimulate the economy but may crowd out private investment through higher interest rates.

The **crowding out effect** occurs when government borrowing increases demand for loanable funds, raising interest rates and reducing private investment. This effect is more significant in closed economies with limited capital inflow.

5.2 Monetary Policy

Core Concept 5.4: Monetary Policy

Monetary policy involves central bank actions controlling the money supply and interest rates.

- **Expansionary monetary policy** decreases interest rates and increases the money supply to stimulate economic activity.
- **Contractionary monetary policy** increases interest rates and reduces the money supply to combat inflation.

The central bank influences rates through open market operations, reserve requirements, and the discount rate.

The central bank influences rates through the following monetary policy tools.

Core Concept 5.5: Monetary Policy Tool (*Open Market Operations*)

Open market operations involve buying and government bonds to affect bank reserves.

- **Purchasing bonds** injects reserves and expand the money supply in the economy
- **Selling bonds** build up reserves and contract the money supply in the economy

Core Concept 5.6: Monetary Policy Tool (*Required Reserve Ratio*)

Changes in the **required reserve ratio (RRR)** affect banks' lending capacity. Lowering the RRR expands the money supply and raising it contracts the money supply. Keep in mind that the ultimate change in money supply is amplified due to the multiplier effect in fractional reserve banking (Core Concept 3.2).

Historically, the RRR was set by the Federal Reserve and publicly announced. However, as of March 2020 (*COVID pandemic*), the Fed reduced the RRR to 0% for all deposits, effectively eliminating it as an active policy tool.

In contrast, some central banks (*such as the People's Bank of China*) continue to actively adjust reserve requirements as a monetary policy tool.

Core Concept 5.7: Monetary Policy Tool (*Discount Rate, Interest on Reserves, and Federal Funds Rate*)

The **discount rate** is the interest rate banks pay to borrow from the central bank. Consequently, no bank would ever rationally borrow from another commercial bank with a higher interest rate than the discount rate. Thus,

lowering the discount rate encourages borrowing because it effectively acts as the ceiling for all other interest rates; raising it has the opposite effect.

Interest on reserves (IOR) is the rate of interest that a central bank pays to commercial banks on the funds they hold in their reserve accounts at the central bank. This effectively sets a floor for interest rates because no lender would ever rationally lend at a rate below the IOR.

Through lowering and raising these two figures, the central bank can effectively manage short-term rates across the economy, influencing the level of economic activity as a result. This target rate range is called the **Federal Reserve Rate (FFR)**.

Figure 18: Federal funds rate (July 1954 - February 2026)

*The FFR serves as the primary benchmark for the global financial system, effectively anchoring the interest rates for all other types of borrowing. To maintain economic stability, the **Federal Open Market Committee (FOMC)** meets eight times annually to evaluate and adjust the FFR target.*

The Federal Reserve operates under a dual mandate, seeking to maintain an economy characterized by price stability—quantified as a 2% long-run inflation target—and maximum employment, which is traditionally associated with a NRU centered around 4%.

6 International Trade

6.1 Balance of Payments

The **Balance of Payments (BOP)** is a comprehensive record of all economic transactions between the residents of a country and the rest of the world over a specific period. It is composed of two sections.

Core Concept 6.1: Current Account

The **current account** measures the net balance of **value** flow into a country in the form of goods, services, income, and transfers. It essentially represents a nation's short-term economic transactions.

A current account deficit means a country is importing more than it exports and is a net borrower from abroad. A surplus indicates the opposite.

It's helpful to think of the current account as a country's shopping bag.

Core Concept 6.2: Capital (Financial) Account

The **capital (financial) account** records the net change in national ownership of assets. It reflects how a country finances the items in its current account or where it invests its current account surplus. When a country imports in a good with the value of 100, there must be an export of assets (*usually currency*) of the same value of 100.

A capital account deficit means a country is gaining more assets than is losing. A surplus indicates the opposite.

It's helpful to think of the capital account as a country's wallet.

In theory, the current account and capital account sum to zero: a current account deficit must be financed by a capital account surplus (*net borrowing from abroad*).

6.2 Exchange Rates

Core Concept 6.3: Exchange Rates

The **exchange rate** represents the price of one currency in terms of another. Currency **appreciation** means a currency becomes more valuable relative to another, and **depreciation** means it becomes less valuable. The following is the foreign exchange market graph for the USD/EUR pair.

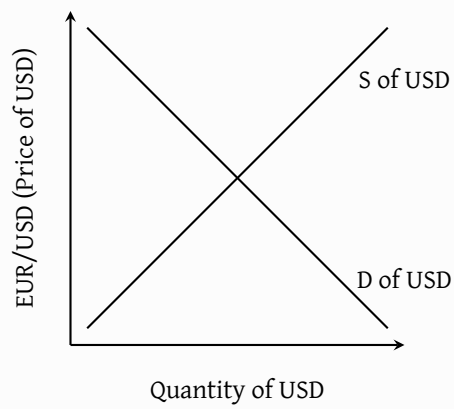

Figure 19: USD/EUR currency market

Like other markets, exchange rates are determined by supply and demand. Factors affecting currency demand include the following.

- Relative interest rates (*higher rates attract investors towards assets that are bought via that currency*)
- Relative price levels (*lower prices increase global economy competitiveness*)
- Relative national income (*higher income increases imports and currency demand*)
- Rariffs (*tariffs reduce import demand*).

Core Concept 6.4: Real Interest Rate Parity

The **Real Interest Rate Parity** condition states that differences in real interest rates between countries drive capital flows. Higher domestic real interest rates relative to foreign rates attract capital inflows, appreciating the domestic currency.

These capital flows continue until real interest rates equalize across countries, implying that in integrated capital markets, real interest rates should converge globally.

In practice, real interest rates do not perfectly converge due to the following. Consequently, the RIRP condition serves rather as a long-run tendency than a constant state of equilibrium.

- **Transaction Costs:** Fees and regulations that impede the free flow of capital.
- **Risk Premiums:** Higher rates required to compensate for political or default risk in specific countries.
- **Non-Tradable Goods:** Differences in local inflation that prevent Purchasing Power Parity (PPP) from holding.